CHARACTERISATIONS OF ABSOLUTELY FLAT RINGS BY WEAK DIMENSION, REDUCEDNESS WITH ALL PRIMES MAXIMAL, AND FIELD LOCAL RINGS

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Editorial note. The six references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Pure ideals and the generation of a finitely generated pure ideal by an idempotent. invoked at proof lines 30265-30268. Is the step that converts absolute flatness into idempotent generation of principal ideals.
- The connection between idempotents and open-and-closed subsets of the spectrum, and the criterion on rings with only minimal primes being a field. used at proof lines 30269-30273. Gives that every prime is maximal and, with nilpotence, that A is reduced.
- A ring all of whose local rings are fields has all modules flat. invoked at proof lines 30283-30285. Closes the cycle by giving (4) implies (2).

Source reference. Original lemma 04PR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6124–6124. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05KK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7204–7204. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 655–655. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00EC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 529–529. Cross-chapter navigation pointer into the

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Source reference. Original lemma 00EU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 548–548. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04MG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6042–6042. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let A be a ring. The following are equivalent*

(1) *A has weak dimension ≤ 0,*
(2) *A is absolutely flat,*
(3) *A is reduced and every prime is maximal, and*
(4) *every local ring of A is a field.*

Proof. The equivalence of (1) and (2) is immediate.

Proof of (1) $\Rightarrow$ (3). Assume A is absolutely flat. Then every ideal of A is pure, see Algebra, Definition 04PR. Hence every finitely generated ideal is generated by an idempotent by Algebra, Lemma 05KK. If $f \in A$, then $(f) = (e)$ for some idempotent $e \in A$ and $D(f) = D(e)$ is open and closed (Algebra, Lemma 00EC). This implies every prime ideal of A is maximal by Algebra, Lemma 04MG. Moreover, if f is nilpotent, then $e = 0$ hence $f = 0$. Thus A is reduced.

Proof of (3) $\Rightarrow$ (4). If A is reduced and every prime is maximal, then every local ring is a field by Algebra, Lemma 00EU.

Proof of (4) $\Rightarrow$ (2). If every local ring of A is a field, then every A-module is flat by Algebra, Lemma 00HT. $\square$